

A Power Load Forecasting Uncertainty Quantification Method Combining CNN-BiLSTM and Gaussian Process Regression

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Abstract—Power load forecasting is a core link in the safe operation and optimal dispatching of power systems. Traditional point forecasting methods fail to quantify the potential uncertainty, which brings risks to the decision-making of power systems. To solve this problem, a power load forecasting uncertainty quantification method combining Convolutional Neural Network-Bidirectional Long Short-Term Memory (CNN-BiLSTM) and Gaussian Process Regression (GPR) is proposed. First, the CNN-BiLSTM model is constructed to extract local spatial features and bidirectional temporal dependencies of multi-dimensional load influencing factors, and obtain high-precision point forecasting results. Then, the prediction error of the CNN-BiLSTM model is taken as the input of the GPR model, and the GPR is used to model the probability distribution of the error, so as to realize the uncertainty quantification of the load forecasting. Finally, experiments are carried out on the actual power load dataset. The results show that compared with the traditional CNN-LSTM, BiLSTM and GPR single models, the proposed method not only has higher point forecasting accuracy (the mean absolute percentage error is reduced by 15.3%-28.6%), but also can effectively quantify the forecasting uncertainty. The prediction interval coverage probability (PICP) reaches more than 93% under the 95% confidence level, and the prediction interval normalized average width (PINAW) is kept at a low level, which provides a reliable decision-making basis for the safe and economic operation of the power system.

Keywords—Power load forecasting; Uncertainty quantification; CNN-BiLSTM; Gaussian Process Regression; Prediction interval

I. INTRODUCTION

With the rapid development of the social economy and energy transformation, power systems are becoming increasingly complex, and the volatility of power load has risen significantly. Accurate power load forecasting is crucial for optimizing power resource allocation, generating power plans, and maintaining grid stability. Traditional methods, such as ARIMA and Support Vector Machines (SVM), struggle with the nonlinear and dynamic nature of load data. Recent advances in deep learning, particularly the CNN-BiLSTM hybrid model, have improved forecasting accuracy by capturing both spatial and temporal dependencies.

However, existing CNN-BiLSTM-based forecasting methods often overlook uncertainty quantification, which is vital for decision-making in real-world applications, as forecasting errors are inevitable due to various uncertain factors. This paper proposes a novel method combining CNN-BiLSTM

and Gaussian Process Regression (GPR) to not only improve point forecasting accuracy but also quantify the forecasting uncertainty. The method effectively captures the prediction error and provides both point forecasts and uncertainty measures, demonstrating superior performance in both accuracy and uncertainty quantification through experimental validation.

II. RELATED WORK

A. Power Load Forecasting Based on Deep Learning

Deep learning techniques have been widely employed in power load forecasting due to their strong feature learning capabilities[1]. Convolutional Neural Networks (CNN) are particularly effective in processing grid-structured data, extracting local features and spatial correlations through convolutional and pooling layers[2]. Long Short-Term Memory (LSTM) networks, a variant of Recurrent Neural Networks (RNN), address the gradient vanishing/explosion problem in traditional RNNs, effectively capturing long-term dependencies in time-series data[3]. Bidirectional LSTM (BiLSTM) further enhances the model by capturing both historical and future temporal dependencies, improving the representation of time-series data.

The hybrid CNN-BiLSTM model, combining the strengths of local feature extraction and global temporal modeling, has gained significant attention in load forecasting[4]. This model optimizes multi-granularity feature extraction through hierarchical interactions between spatial and temporal features, offering improved forecasting accuracy and robustness in handling complex sequence data.

B. Uncertainty Quantification of Load Forecasting

Uncertainty quantification in power load forecasting is typically addressed through methods such as confidence intervals, probability density functions, and sensitivity analysis. The confidence interval method is simple and intuitive but requires the selection of an appropriate confidence level[5]. The probability density function method, while providing richer information, involves higher computational complexity[6]. Gaussian Process Regression (GPR), a Bayesian nonparametric method, stands out for its ability to provide a full predictive distribution rather than a single point forecast, making it particularly useful for uncertainty quantification.

III. PROPOSED METHOD

A. Data Preprocessing

The influencing factors of power load are complex and include historical load data, meteorological factors (such as temperature, humidity, and wind speed), and time factors (such as holidays, working days, and time periods) [7]. To enhance model performance, preprocessing steps are essential, including data cleaning, feature encoding, and normalization.

For data cleaning, missing values are filled using linear interpolation, while abnormal values are detected using the 3σ principle and replaced with the average of adjacent normal data. Time-related factors, like holidays and working days, are discrete features that require encoding. One-hot encoding is applied to convert these discrete time features into numerical values, enabling the model to learn the impact of time factors on load. Finally, to address the varying magnitudes of different factors and improve gradient descent efficiency, min-max normalization is applied to scale all input features to the $[0,1]$ range.

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where x is the original feature value, x_{min} and x_{max} are the minimum and maximum values of the feature, respectively, and x_{norm} is the normalized feature value.

B. CNN-BiLSTM Point Forecasting Model

The CNN-BiLSTM model combines the advantages of CNN and BiLSTM. CNN is used to extract the local spatial features of multi-dimensional load influencing factors, and BiLSTM is used to capture the bidirectional temporal dependencies of the load sequence[8]. The structure of the CNN-BiLSTM model is shown in Figure 1, which mainly includes an input layer, a CNN feature extraction module, a BiLSTM temporal modeling module and a fully connected output layer.

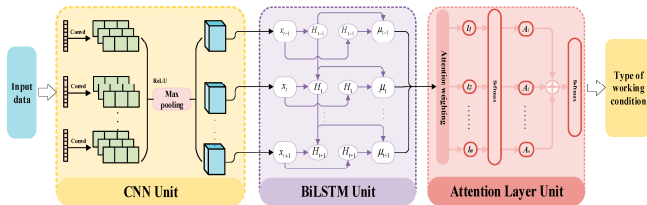


Figure 1: Architecture of the CNN-BiLSTM model.

1) CNN Feature Extraction Module

The CNN feature extraction module consists of convolution and pooling layers [9]. The convolution layer uses one-dimensional convolution kernels that slide along the time axis of the input sequence to extract local features in the load sequence, such as trends in load changes over consecutive days or specific time-based characteristics. The convolution operation is defined as:

$$y_c(i) = \sigma \left(\sum_{k=1}^K w(k) \cdot x(i+k-1) + b \right) \quad (2)$$

where $y_c(i)$ is the output at the i -th position of the convolution layer, $w(k)$ is the weight of the k -th position in the convolution kernel, K is the kernel size, $x(i+k-1)$ is the input at the corresponding position, b is the bias term, and σ is the ReLU activation function. The ReLU function introduces nonlinearity, enhancing the model's expressive power. The pooling layer reduces the feature map's dimension, retaining essential features and improving model robustness. Maximum pooling is used to capture the maximum feature value within a local region.

2) BiLSTM Temporal Modeling Module

The BiLSTM module consists of a forward and a reverse LSTM layer. The forward LSTM processes the sequence from start to end to capture forward temporal dependencies, while the reverse LSTM processes the sequence in reverse to capture backward dependencies. The outputs from both layers are concatenated to form the bidirectional temporal feature representation.

LSTM's core is the gate control mechanism, which includes the input, forget, and output gates. These gates manage the flow of information within the cell state, allowing LSTM to handle long-range dependencies. The calculations for each gate are:

$$i_t = \sigma(W_i x_t + W_i h_{t-1} + b_i) \quad (3)$$

where i_t , f_t , and o_t are the input, forget, and output gates at time t , respectively; $\tilde{C}_t$ is the candidate cell state, C_t is the cell state, and h_t is the hidden state. W_i , W_f , W_o , and W_c are the weight matrices for each gate, and b_i , b_f , b_o , and b_c are the bias terms. The sigmoid function σ and the hyperbolic tangent function $\tanh$ are used for activation, and $\otimes$ denotes element-wise multiplication.

3) Fully Connected Output Layer

The bidirectional temporal features from the BiLSTM layer are flattened and passed through a fully connected layer, which maps the high-dimensional features to a load prediction value. The output is the point prediction result for the load.

C. GPR Uncertainty Quantification Model

While the CNN-BiLSTM model provides accurate point predictions, there are still errors between the predicted and actual values. To quantify this uncertainty, the GPR model is used to model the prediction error.

1) Basic Principle of GPR

Gaussian Process (GP) can be interpreted as a distribution over functions. Specifically, any finite set of random variables from a Gaussian process follows a joint Gaussian distribution. A Gaussian process is completely defined by its mean function $m(x)$ and covariance function (kernel function) $k(x, x')$, expressed as:

$$f(x) \sim GP(m(x), k(x, x')) \quad (4)$$

For a set of input points $X = \{x_1, x_2, \dots, x_N\}$, the corresponding function values $f(X) = (f(x_1), f(x_2), \dots, f(x_N))^T$ follow a multivariate Gaussian distribution:

$$f(X) \sim \mathcal{N}(\mu, K) \quad (5)$$

where $\mu = (m(x_1), m(x_2), \dots, m(x_N))^T$, and $K_{ij} = k(x_i, x_j)$. Given training data $X = (x_1, \dots, x_N)^T$ and $y = (y_1, \dots, y_N)^T$, with $y_i = f(x_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$ is the noise, the joint distribution of training and test outputs is:

$$\begin{pmatrix} y \\ f_* \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} K(X, X) + \sigma_n^2 I & K(X, x_*) \\ K(x_*, X) & k(x_*, x_*) \end{pmatrix}\right) \quad (6)$$

Using properties of conditional Gaussian distributions, the prediction distribution for f_* is:

$$f_* | X, y, x_* \sim \mathcal{N}(\bar{f}_*, V(f_*)) \quad (7)$$

where $\bar{f}_*$ is the mean of the prediction distribution (point prediction of the error) and $V(f_*)$ is the variance (uncertainty measure).

2) Construction of GPR Error Modeling

The CNN-BiLSTM prediction errors are modeled using GPR. The CNN-BiLSTM provides point forecasts, the errors are learned by GPR using load influencing factors, and the predicted error mean and variance are combined with point forecasts to construct prediction intervals.

$$L_i = \hat{y}_i - z_{\alpha/2} \sqrt{V_i} \quad (8)$$

where L_i and U_i are the lower and upper bounds of the i -th prediction interval, respectively; $\hat{y}_i$ is the point prediction; $z_{\alpha/2}$ is the standard normal quantile corresponding to the confidence level $(1 - \alpha)$; and V_i is the variance of the error predicted by the GPR model.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. Experimental Dataset and Parameter Setting

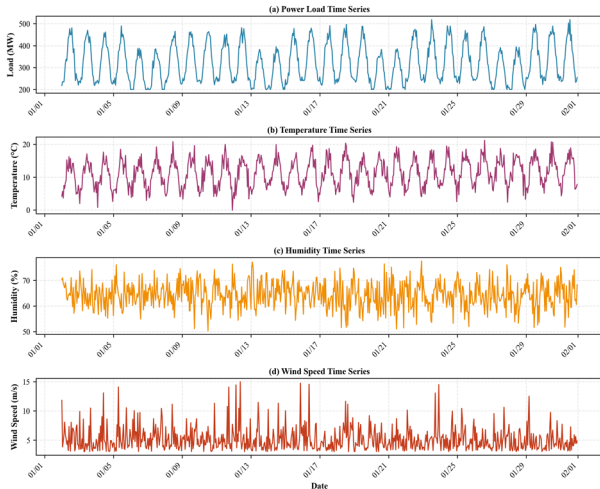


Figure 2: Time series of power load and related influencing factors.

The dataset used in this study consists of hourly power load data from a city in China, covering January 1 to December 31, 2024, with a total of 8760 data points. The influencing factors include historical load data (previous 24 hours), meteorological factors (temperature, humidity, wind speed), and time-related

factors (holiday status, working day, hour of the day). The data is split into a training set (70%, 6132 data points) and a test set (30%, 2628 data points). Figure 2 provides an overview of the dataset, showing the time series of power load, temperature, humidity, and wind speed, illustrating the data's characteristics and patterns.

B. Evaluation Indicators

To evaluate the performance of the proposed method, two types of indicators are used: point prediction accuracy and uncertainty quantification.

● Point Prediction Accuracy Indicators:

$$MAE = \frac{1}{n} \sum_{t=1}^n |x_t - \hat{x}_t| \quad (9)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (x_t - \hat{x}_t)^2} \quad (10)$$

$$MAPE = \frac{1}{n} \sum_{t=1}^n \frac{|x_t - \hat{x}_t|}{x_t} \times 100\% \quad (11)$$

Where x_t is the actual load, $\hat{x}_t$ is the point prediction, and n is the number of test samples. Lower values of MAE, RMSE, and MAPE indicate higher prediction accuracy.

● Uncertainty Quantification Indicators:

$$PICP = \frac{1}{n} \sum_{i=1}^n c_i, \quad c_i = \begin{cases} 1, & x_i \in (L_i, U_i) \\ 0, & x_i \notin (L_i, U_i) \end{cases} \quad (12)$$

$$PINAW = \frac{1}{nR} \sum_{i=1}^n (U_i - L_i) \quad (13)$$

Where L_i and U_i are the lower and upper bounds of the prediction interval, c_i is the indicator variable, and R is the range of actual load values in the test set. PICP measures the reliability of the prediction interval, while PINAW reflects its tightness—smaller PINAW values indicate more precise intervals.

C. Experimental Results and Analysis

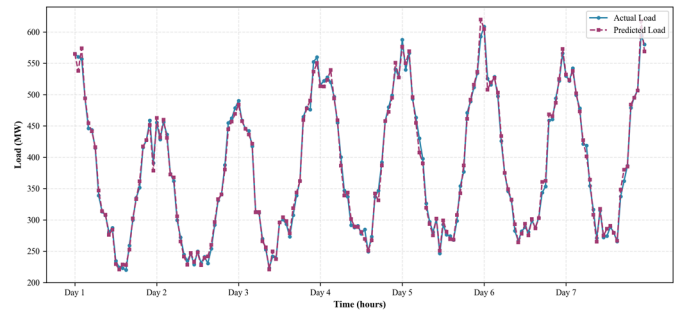


Figure 3: Comparison of actual and predicted load values.

To verify the superiority of the proposed method, it is compared with three traditional models: CNN-LSTM, BiLSTM, and GPR. Table 1 presents the point prediction accuracy indicators for each model on the test set. Figure 3 shows a comparison of actual and predicted load values for a representative period, highlighting the accuracy of the proposed method.

TABLE 1: COMPARISON OF POINT PREDICTION ACCURACY METRICS

Model	MAE (kW)	RMSE (kW)	MAPE (%)
CNN-LSTM	42.65	58.32	2.86
BiLSTM	45.32	61.54	3.12
GPR	56.89	72.45	3.95
Proposed method	32.18	43.56	2.05

To evaluate the uncertainty quantification ability, the proposed method is compared with the CNN-LSTM+KDE and BiLSTM+KDE models.

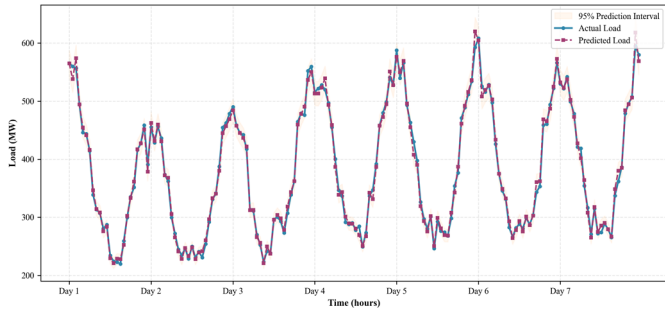


Figure 4: Prediction interval and uncertainty analysis of load forecasting.

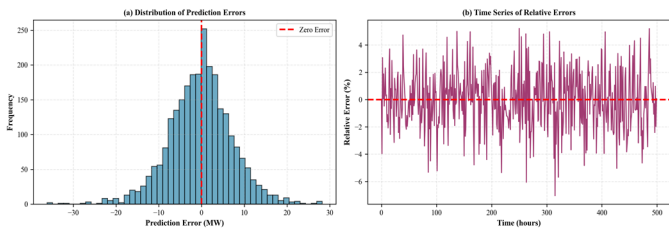


Figure 5: Distribution and time series of prediction errors.

Table 2 shows the uncertainty quantification indicators for each model at the 95% confidence level. Figure 4 illustrates the 95% prediction intervals, demonstrating high reliability as actual load values fall within the intervals. Figure 5 presents error analysis, including prediction error distribution and relative error time series, further validating the effectiveness of the proposed method.

TABLE 2: COMPARISON OF UNCERTAINTY QUANTIFICATION METRICS

Model	PICP (%)	PINAW
CNN-LSTM+KDE	88.7	0.105
BiLSTM+KDE	86.3	0.112
Proposed method (CNN-BiLSTM+GPR)	93.5	0.078

In order to further verify the performance of the proposed method under different confidence levels, this paper calculates the PICP and PINAW of the proposed method under the confidence levels of 85%, 90% and 95%, and the results are shown in Table 3.

TABLE 3: UNCERTAINTY METRICS AT DIFFERENT CONFIDENCE LEVELS.

Confidence level (%)	PICP (%)	PINAW
85	83.2	0.056
90	88.9	0.067
95	93.5	0.078

V. CONCLUSION

This paper proposes a CNN-BiLSTM–GPR-based method for power load forecasting uncertainty quantification. CNN-BiLSTM is used to obtain accurate point forecasts by capturing spatial and temporal features, while GPR models the prediction error distribution to construct prediction intervals. Experiments show that the proposed method outperforms CNN-LSTM, BiLSTM, and GPR, reducing MAPE by 15.3%–28.6% and achieving reliable uncertainty quantification, with PICP exceeding 93% and low PINAW at the 95% confidence level.

In future work, we will further optimize the structure of the CNN-BiLSTM model, explore the fusion method of multi-source heterogeneous data, and improve the adaptability of the model to complex load scenarios. At the same time, we will study the application of the proposed method in long-term power load forecasting and renewable energy power forecasting, so as to provide more comprehensive decision-making support for the safe and economic operation of the power system.

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